Enhancing Visual Language Models: Evaluating and Improving Negation Comprehension in Medical Imaging

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1 Research question and motivation

How well do current vision-language models (VLMs) understand negation in the context of medical imaging, and how can their performance be improved for negation-sensitive tasks such as retrieval and classification?

Accurate interpretation of negation in medical imaging is essential to avoid misdiagnoses and ensure patient safety. VLMs, such as CLIP, have demonstrated strong performance across multimodal tasks, including medical diagnostics. However, recent research highlights their critical limitation in comprehending negation, often conflating affirmative and negated clinical statements.

Proper understanding of negation, such as distinguishing "no evidence of pneumonia" from "evidence of pneumonia," directly impacts clinical decisions. The CheXpert [2] dataset, containing chest radiographs annotated with affirmative and negated clinical conditions, provides a relevant benchmark to explore this limitation.

Alhamoud et al.'s NegBench [1] shows CLIP-based VLMs struggle with negation (e.g., 24.6-33.2 percent accuracy drops on CheXpert), proposing synthetic data fine-tuning as a fix (10 percent retrieval, 40 percent MCQ gains). While medical VLMs like BioMedCLIP improve general tasks, negation remains a weakness. This project benchmarks VLM performance on medical imaging datasets starting with CheXpert. We aim to rigorously analyze their negation failures and propose data-driven approaches, including fine-tuning on synthetic negation-enriched datasets, to enhance their performance. Ultimately, the goal is to improve VLM robustness, ensuring accurate and reliable applications in medical imaging.

2 Data sources

Our primary source of data will be the CheXpert-v1.0-small dataset on Kaggle, which is a downsized version of the original CheXpert. This dataset includes 224,316 chest radiographs from 65,240 patients and has already been optimized and processed as it features frontal and lateral views of chest radiographs. It also comes with uncertainty labels along with evaluation sets that have been modified by radiologists. The dataset features pictures in jpg format along with 18 additional points of classification for each given image, such as age and the presence of various conditions.

We intend to work with another 1-3 medical datasets to demonstrate the state of negation-related inquiries within VLMs and consider the potential impact of these models within medicine. Additional datasets provide for more robust testing, as we can more critically assess the performance of base and modified VLM models to determine how prominent the issue of negation is and by what accuracy these models can be improved.

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Similar to CheXpert, we intend to source medical datasets by analyzing them across other medical papers to find high-value medical issues to investigate. This will limit the amount of data cleaning required, as datasets sourced this way will already have been cleaned and provide explicit classes on which negation can be performed with. The initial processing of the data will be to identify the accuracy of negation within existing VLMs such as CLIP. Potential challenges will likely arise in formatting, as different datasets may follow different image formats and types, such as jpg and png, which we plan on processing differently following a robust analysis of each dataset format.

3 Methodology and analysis plan

Firstly, for each dataset, we intend on performing Exploratory Data Analysis, to determine the distribution of each class within each dataset, for example the proportion of a sample that has condition X. This will help provide a rough idea of how common or rare a condition is, which will also provide context on the relationship between conditions, which will be helpful to in assessing responses to negation queries.

We next construct benchmarks for each dataset, such as the CheXpert dataset for radiographs to evaluate the normal performance of several VLMs, like CLIP, to provide insight on base-level tasks that don't include negation. Additionally, a diagnostic task will be established to assess baseline negation comprehension accuracy. Existing VLMs, such as CLIP, will be benchmarked using GPU resources provided by Rutgers University.

Ideally, we will look for a base performance accuracy of 90 percent or higher on non-negation-related queries before we proceed to consider negation-based queries. This will serve as a prerequisite to ensure meaningful assessment.

Following this baseline evaluation, models will undergo fine-tuning on synthetic, negation-enriched data to improve their ability to handle negation. The effectiveness of these methods will be tested using metrics like accuracy and recall, specifically comparing affirmative versus negated statements.

We will primarily leverage Python, including data science and artificial intelligence-related dependencies such as Pandas, NumPy, HuggingFace Transformers, and PyTorch to facilitate dataset construction, modeling, and evaluation. Additionally, as scaling VLM testing and fine-tuning will become quite computationally intensive, we intend to leverage research-grade GPUs to efficiently benchmark existing VLM models and experiment with fine-tuning to improve responses to negation-based tasks.

4 Expected outcomes and evaluation

We expect to find that even state-of-the-art VLM models have a large accuracy gap between non-negation and negation-related tasks, which will add challenges to condition identification and other areas within medicine. We intend on testing alternative methods to improve accuracy on negation tasks, such as contrastive learning, and comparing accuracies to regular VLM models to determine if alternative architectures improve performance.

References

[1] K. Alhamoud, S. Alshammari, Y. Tian, G. Li, P. H. S. Torr, Y. Kim, and M. Ghassemi. Vision-Language Models Do Not Understand Negation. *arXiv preprint arXiv:2501.09425*, 2025. `https://arxiv.org/abs/2501.09425`.

[2] J. Irvin et al., CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. *arXiv preprint arXiv:1901.07031*, 2019. [Online]. Available: `https://arxiv.org/abs/1901.07031`